Gold-frame calibration — Experiment plan

ANTARES beamline (SOLEIL) · calibration of the sample-stage model with SPEM lines over the gold reference frame · plan for the beamtime, November 2026

Contents

1. Goal
2. What is needed
3. The measurement at a glance
4. Before the beamtime
5. Procedure at the beamline
6. The SZ lines
7. Optional: SN points at other SRz
8. Analysis and acceptance
9. If something goes wrong
10. Logbook sheet
11. Background: why SZ lines over a diamond

Goal

Measure the constants of the stage model used by QuickAlignment, so that samples are found from their position on the plate at any SRn / SRz:

constant	meaning
ST0, SZ0	ST, SZ that put the plate centre on the beam (SRn 120, SRz 0)
SN_f	SN that puts the riser top of the standard plate in the analyzer focal plane
SRn axis (x, y)	where the SRn rotation axis pierces the plate
SRz axis offsets (along ST, depth)	where the SRz rotation axis is, relative to the plate surface

The fit also gives the in-plane rotation of the frame, the tilt of its gold face and the ST / SZ motor scales, for the record. The result is a new configuration file in `configs/` that QuickAlignment loads at its next start.

Expected accuracy (simulations, see the report): about 1–2 µm for ST0, SZ0 and the SRn axis, about 5–20 µm for the SRz axis.

Method in one sentence: only SZ is scanned (fly scans); with the frame turned by 45° (SRn 165 / 75) its edges are diagonal, so where an SZ line crosses them also gives the horizontal position, and every edge of every line is fitted at once.

What is needed

Sample. The gold reference frame (2 × 2 mm outside, 1 × 1 mm hole) glued at the centre of the riser of a **standard Ta plate**. If it is known to be off-centre, note its position (x, y in mm on the plate, front view, tab up): it is entered in the program. The frame is taken as the plate reference: an error in where it is glued goes into ST0, SZ0 and the axes.

Beamline.

- Focusing optic aligned and focused (capillary or zone plate + OSA). The configuration is saved for that optic.
- Total-intensity SPEM signal (`heason/total_int`, or the MBS `image32` summed): no angle or energy resolution needed.
- SZ fly scans with steps down to 0.02 mm; ST, SN, SRn, SRz moved between lines.
- Motor ranges used by the program: ST −9 … 9 mm, SN −30 … 65 mm, SZ 35 … 45 mm.

Software. `GoldFrameCalibration/GoldFrameCalibration.py` (double-click; Windows: `.pyw`) on a PC with the repository and `pip install -r requirements.txt` (PyQt5, numpy, matplotlib, h5py). It reads the ANTARES `.nxs` files directly.

People. One person at the beamline control, one checking the scans and keeping the logbook.

The measurement at a glance

step	what	gives	time (estimate)
0	mount the plate, SRn 120, SRz 0, PIX = PIY = 150 µm		15 min
1	overview map ST × SZ, ±2 mm around the frame	where the frame is; start of the fit	15–20 min
2	SN scans (ST following) on a gold bar and on the riser; optic focused on the gold	SN_gold, gold height *h*, analyzer acceptance, optic reference	15 min
3	make the plan in the program (33 SZ lines in 9 settings), check it	the positions of every line	5 min
4	measure the 33 SZ lines (groups A, B, C)	the data of the fit	30–40 min
5	optional: SN points at 6 SRz	cross-check of the SRz axis depth	30 min
6	fit, check, save	the configuration	10 min
7	try the configuration on a sample of known position at another rotation	verification	15 min

Times are rough planning figures, assuming about 0.1 s per point in the SZ fly scans and 1–2 min per SRn / SRz change. Total: about 2 h, 2.5 h with the optional SN points.

Before the beamtime

1. Check that the program starts and reads a file: *Load a simulated example* in the GoldFrameCalibration window simulates the whole measurement, loads it and fits it (no beam needed). Try *Fit* and *Save* on a copy of `configs/` if you want to rehearse.
2. Decide which optic is used and make sure the newest configuration of that optic is in `configs/`: the lines are placed with it. If the motors were re-homed since, the lines may be 0.5–1 mm off: plan for an extra first fit (see If something goes wrong).
3. Print the logbook sheet and the lines table (the numbers will be made again on the day).

Procedure at the beamline

Step 0 — Mount

- Mount the standard plate with the gold frame, tab up. SRn 120, SRz 0, PIX = PIY = 150 µm.
- Bring the frame onto the beam with ST / SZ (camera), SN so that the gold surface is on the "SN = 0" line of the camera.
- Note the optic in use and its position.

Step 1 — Overview map

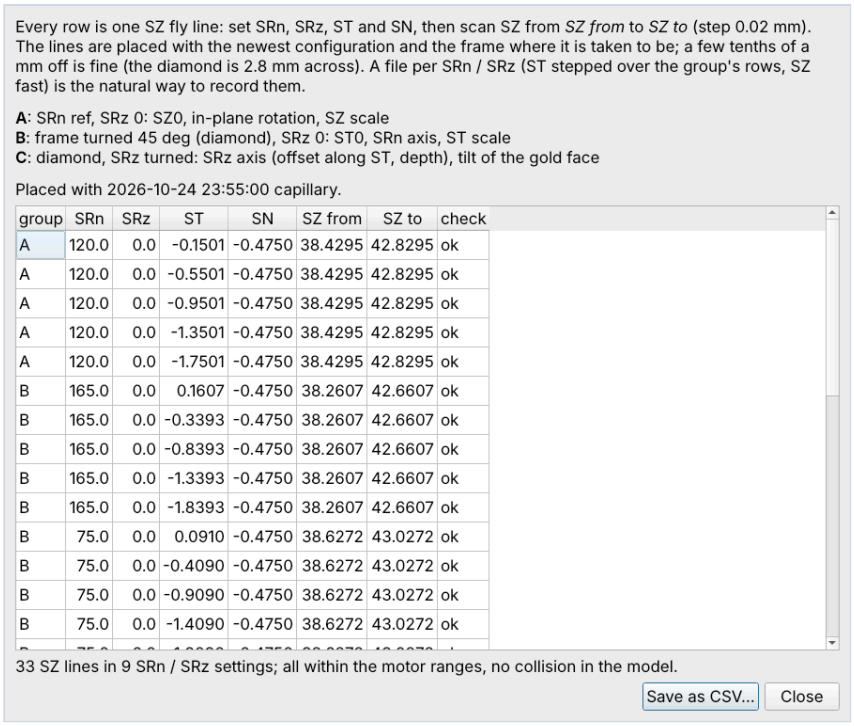
- 2-D map: **ST stepped** (±2 mm around the frame, steps 0.1 mm), **SZ fast** (fly, ±2 mm, step 0.02 mm).
- Check: the whole frame (both bars around the hole) is inside the map. If not, re-centre and repeat.
- Save the file; it is loaded in the fit with the others (it gives the fit its start).

Step 2 — SN scans and optic focus

- On a **gold bar** (e.g. the right bar), an **SN scan with ST following** (ΔST = −ΔSN, so the point stays on the beam and only slides along it): the intensity peak is **SN_gold**, the SN with the gold top at the analyzer focus. Its width is the analyzer acceptance across its axis.
- The same scan on the bare **riser** next to the frame: the difference of the two peaks is the **gold height *h*** (gold + glue above the riser top; nominal 0.05 mm).
- With the gold at SN_gold, **focus the optic** on the gold and note its position (it becomes the optic reference of the configuration; the program also reads it from a file recorded at SRn 120 / SRz 0).
- Write SN_gold, *h*, the peak width and the optic position in the logbook.

Step 3 — Make the plan

- Open **GoldFrameCalibration**, choose the optic, set the gold height *h* (and the frame position if not at the centre), press **Measurement plan…**.
- Every line must show **ok** (within the motor ranges, no collision in the anti-collision model). **Save as CSV…** for the beamline.
- The lines only need to pass within a few tenths of a mm of where planned: the diamond is 2.8 mm across.



Every row is one SZ fly line: set SRn, SRz, ST and SN, then scan SZ from *SZ from* to *SZ to* (step 0.02 mm). The lines are placed with the newest configuration and the frame where it is taken to be; a few tenths of a mm off is fine (the diamond is 2.8 mm across). A file per SRn / SRz (ST stepped over the group's rows, SZ fast) is the natural way to record them.

A: SRn ref, SRz 0: SZ0, in-plane rotation, SZ scale
B: frame turned 45 deg (diamond), SRz 0: ST0, SRn axis, ST scale
C: diamond, SRz turned: SRz axis (offset along ST, depth), tilt of the gold face

Placed with 2026-10-24 23:55:00 capillary.

group	SRn	SRz	ST	SN	SZ from	SZ to	check
A	120.0	0.0	-0.1501	-0.4750	38.4295	42.8295	ok
A	120.0	0.0	-0.5501	-0.4750	38.4295	42.8295	ok
A	120.0	0.0	-0.9501	-0.4750	38.4295	42.8295	ok
A	120.0	0.0	-1.3501	-0.4750	38.4295	42.8295	ok
A	120.0	0.0	-1.7501	-0.4750	38.4295	42.8295	ok
B	165.0	0.0	0.1607	-0.4750	38.2607	42.6607	ok
B	165.0	0.0	-0.3393	-0.4750	38.2607	42.6607	ok
B	165.0	0.0	-0.8393	-0.4750	38.2607	42.6607	ok
B	165.0	0.0	-1.3393	-0.4750	38.2607	42.6607	ok
B	165.0	0.0	-1.8393	-0.4750	38.2607	42.6607	ok
B	75.0	0.0	0.0910	-0.4750	38.6272	43.0272	ok
B	75.0	0.0	-0.4090	-0.4750	38.6272	43.0272	ok
B	75.0	0.0	-0.9090	-0.4750	38.6272	43.0272	ok
B	75.0	0.0	-1.4090	-0.4750	38.6272	43.0272	ok

33 SZ lines in 9 SRn / SRz settings; all within the motor ranges, no collision in the model.

Save as CSV... Close

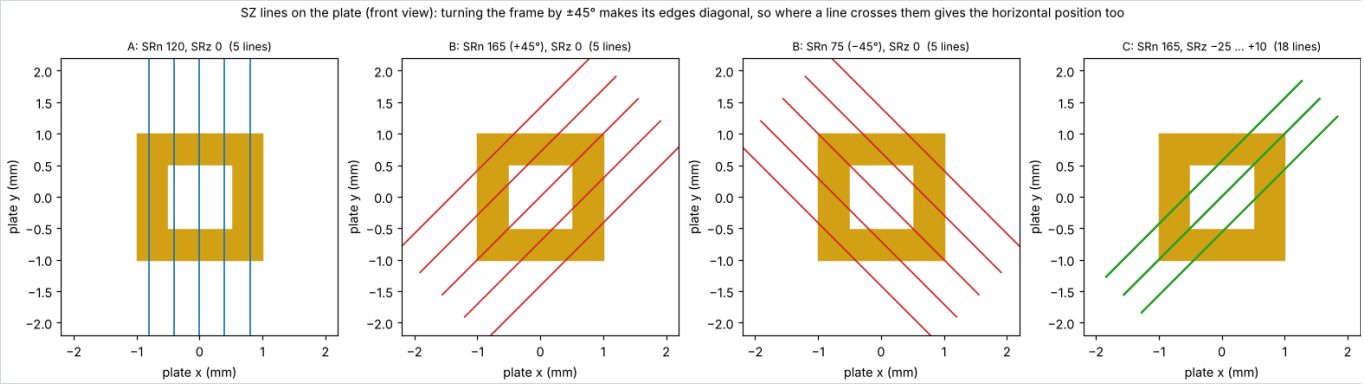
Step 4 — Measure the SZ lines

For each setting S1 … S9 of the settings table, in this order (SRn moves only 120 → 75 → 165, then SRz steps):

1. Set **SRn**, **SRz** and **SN** of the setting (SN is set once per setting: the frame centre is then at the analyzer focus).
2. For each line of the setting, set **ST** and **fly-scan SZ** over the given range with step 0.02 mm. The natural way is one 2-D scan per setting with ST stepped over the setting's ST values; single-line files work as well.
3. Write the file name(s) in the logbook. Look at the data: each line should cross the gold (two or four edges).

group	settings	what it mainly gives
A	SRn 120, SRz 0 (5 lines, ST ±0.8 mm around the centre)	SZ0, in-plane rotation, SZ scale
B	SRn 75 and 165, SRz 0 (5 lines each, ±1 mm across the diamond)	ST0, SRn axis, ST scale
C	SRn 165 at SRz −25, −18, −11, −4, +3, +10 (3 lines each, ±0.4 mm)	SRz axis (along ST, depth), tilt of the gold face

Watch the camera when SRz moves: the program checks every line against its anti-collision model (analyzer and optic solids), but the model is simple.



Step 5 — Optional SN points

See Optional: SN points at other SRz. They cross-check the SRz axis depth as the analyzer sees it; they do not replace group C. They are at SRn 120: measure them right after setting S1 (before SRn moves) or at the end.

Step 6 — Fit and save

In the GoldFrameCalibration window (can be done right after the measurement, at the beamline):

1. **Add files…**: all files, the overview included.
2. **SN gold**: the value of step 2 (by default the SN of a file at SRn 120 / SRz 0). Gold height *h* as measured.
3. *SN at other SRz (optional)…*: enter the SN points if measured.
4. **Fit**, then check the acceptance criteria.
5. **Save as new configuration** with a comment (date, optic, who, files). The file `configs/calibration_<date>_<time>_<optic>.json` is loaded by QuickAlignment at its next start with that optic.

From a console, the same: `python GoldFrameCalibration.py fit FILES… --sn-gold SN --height h --optic capillary --plot fit.png --save`.

Step 7 — Verify

Mount (or use) a sample whose position on the plate is known. In QuickAlignment, go to its predicted position at a rotation other than the calibration ones (for example SRn 100, SRz −15): it should be on the beam, in focus, within about 0.05 mm. If it is not, use *Found here* and note the shift: if several samples show the same shift, the frame is probably not where it was assumed (see If something goes wrong).

The SZ lines

Every line: **SZ fly scan, step 0.02 mm**, over the range given; ST and SN as given; PIX = PIY = 150 µm. Make the plan again in the program on the day (*Measurement plan…*): the numbers below are an example and change with the configuration in use.

Example placed with the configuration **2026-10-24 23:55:00 capillary** (`calibration_2026-10-24_235500_capillary.json`), frame at the riser centre, gold 0.05 mm above the riser top.

setting	group	SRn (°)	SRz (°)	SN (mm)	lines	ST of the lines (mm)	SZ scan (mm)
S1	A	120	0	-0.475	5	-0.150, -0.550, -0.950, -1.350, -1.750	38.429 → 42.830
S2	B	75	0	-0.475	5	0.091, -0.409, -0.909, -1.409, -1.909	38.627 → 43.027
S3	B	165	0	-0.475	5	0.161, -0.339, -0.839, -1.339, -1.839	38.261 → 42.661
S4	C	165	-25	-0.771	3	-1.411, -1.943, -2.474	38.261 → 42.661
S5	C	165	-18	-0.640	3	-1.144, -1.649, -2.153	38.261 → 42.661
S6	C	165	-11	-0.546	3	-0.871, -1.340, -1.809	38.261 → 42.661
S7	C	165	-4	-0.489	3	-0.596, -1.023, -1.450	38.261 → 42.661
S8	C	165	+3	-0.473	3	-0.323, -0.701, -1.080	38.261 → 42.661
S9	C	165	+10	-0.495	3	-0.055, -0.380, -0.704	38.261 → 42.661

All 33 lines (with the check of the program; the last column is for the logbook):

line	group	SRn	SRz	ST	SN	SZ from	SZ to	check	file / done
S1.1	A	120	0	-0.1501	-0.4750	38.4295	42.8295	ok	
S1.2	A	120	0	-0.5501	-0.4750	38.4295	42.8295	ok	
S1.3	A	120	0	-0.9501	-0.4750	38.4295	42.8295	ok	
S1.4	A	120	0	-1.3501	-0.4750	38.4295	42.8295	ok	
S1.5	A	120	0	-1.7501	-0.4750	38.4295	42.8295	ok	
S2.1	B	75	0	0.0910	-0.4750	38.6272	43.0272	ok	
S2.2	B	75	0	-0.4090	-0.4750	38.6272	43.0272	ok	
S2.3	B	75	0	-0.9090	-0.4750	38.6272	43.0272	ok	
S2.4	B	75	0	-1.4090	-0.4750	38.6272	43.0272	ok	
S2.5	B	75	0	-1.9090	-0.4750	38.6272	43.0272	ok	
S3.1	B	165	0	0.1607	-0.4750	38.2607	42.6607	ok	
S3.2	B	165	0	-0.3393	-0.4750	38.2607	42.6607	ok	
S3.3	B	165	0	-0.8393	-0.4750	38.2607	42.6607	ok	
S3.4	B	165	0	-1.3393	-0.4750	38.2607	42.6607	ok	
S3.5	B	165	0	-1.8393	-0.4750	38.2607	42.6607	ok	
S4.1	C	165	-25	-1.4113	-0.7710	38.2607	42.6607	ok	
S4.2	C	165	-25	-1.9429	-0.7710	38.2607	42.6607	ok	

line	group	SRn	SRz	ST	SN	SZ from	SZ to	check	file / done
S4.3	C	165	-25	-2.4745	-0.7710	38.2607	42.6607	ok	
S5.1	C	165	-18	-1.1445	-0.6398	38.2607	42.6607	ok	
S5.2	C	165	-18	-1.6485	-0.6398	38.2607	42.6607	ok	
S5.3	C	165	-18	-2.1525	-0.6398	38.2607	42.6607	ok	
S6.1	C	165	-11	-0.8714	-0.5455	38.2607	42.6607	ok	
S6.2	C	165	-11	-1.3403	-0.5455	38.2607	42.6607	ok	
S6.3	C	165	-11	-1.8093	-0.5455	38.2607	42.6607	ok	
S7.1	C	165	-4	-0.5961	-0.4895	38.2607	42.6607	ok	
S7.2	C	165	-4	-1.0230	-0.4895	38.2607	42.6607	ok	
S7.3	C	165	-4	-1.4499	-0.4895	38.2607	42.6607	ok	
S8.1	C	165	+3	-0.3226	-0.4726	38.2607	42.6607	ok	
S8.2	C	165	+3	-0.7012	-0.4726	38.2607	42.6607	ok	
S8.3	C	165	+3	-1.0797	-0.4726	38.2607	42.6607	ok	
S9.1	C	165	+10	-0.0552	-0.4950	38.2607	42.6607	ok	
S9.2	C	165	+10	-0.3797	-0.4950	38.2607	42.6607	ok	
S9.3	C	165	+10	-0.7041	-0.4950	38.2607	42.6607	ok	

Optional: SN points at other SRz

At each row: set SRn, SRz, ST and SZ, then an **SN scan with ST following** (ΔST = −ΔSN) on the frame's **right bar**; note the **ST and SN at the intensity peak**. Enter them in *SN at other SRz (optional)…* (or a CSV with the columns `srn, srz, st, sz, sn`). Rows without a measured SN are not used. The expected SN is where the newest configuration puts the peak.

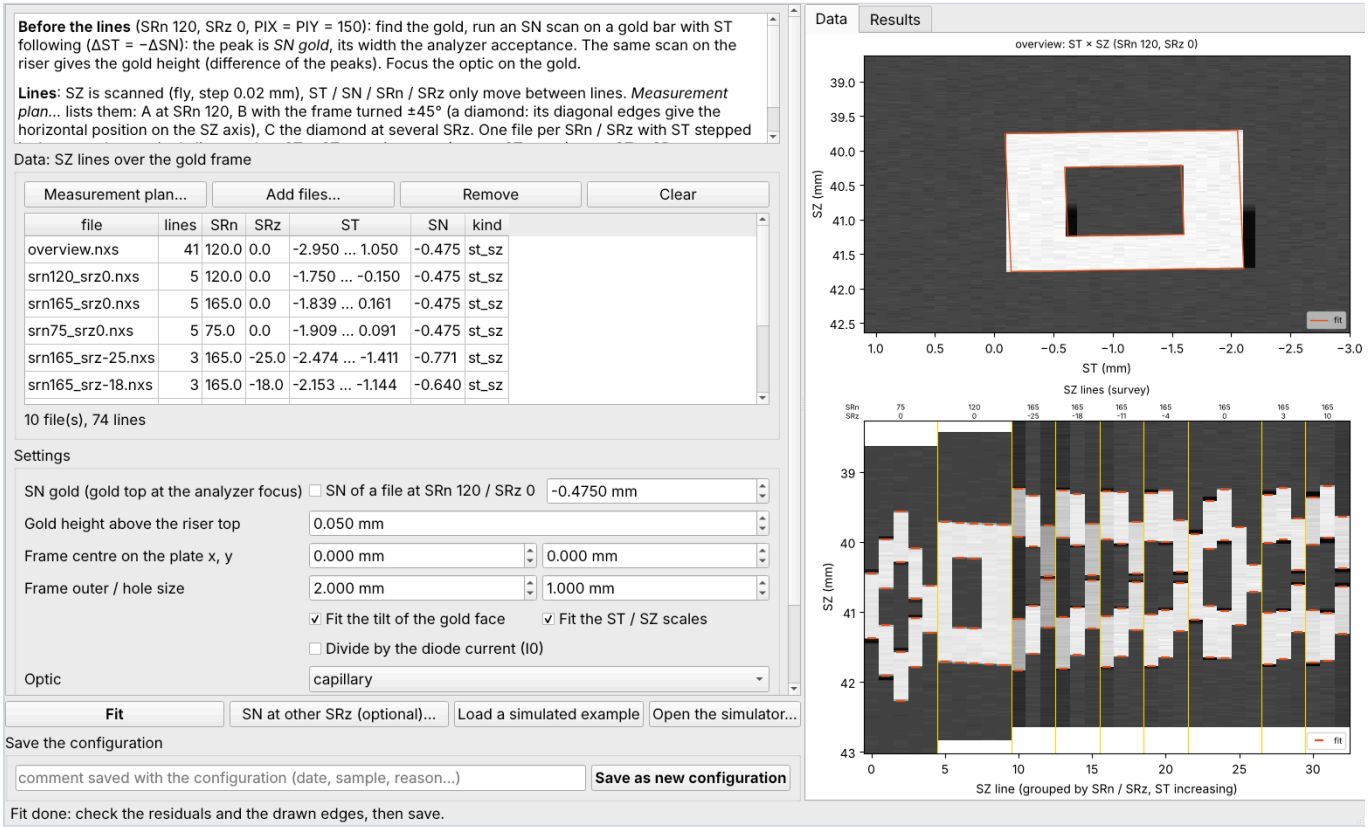
point	SRn	SRz	ST start	SZ	expected SN	measured ST	measured SN
P1	120	-25	-2.7230	40.6295	-1.1347		
P2	120	-18	-2.4672	40.6295	-0.9058		
P3	120	-11	-2.1853	40.6295	-0.7098		
P4	120	-4	-1.8817	40.6295	-0.5495		
P5	120	+3	-1.5608	40.6295	-0.4275		
P6	120	+10	-1.2274	40.6295	-0.3455		

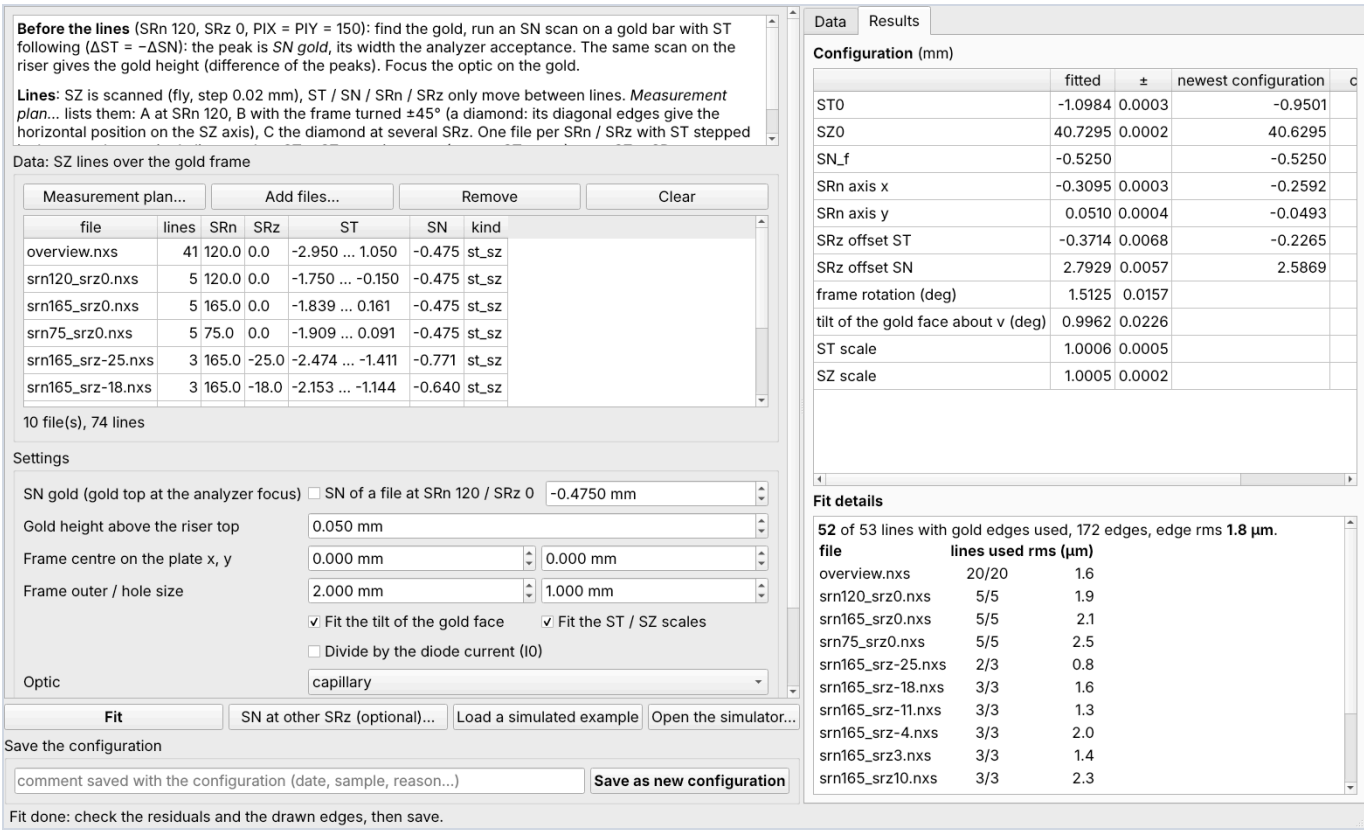
Analysis and acceptance

After *Fit*, the *Results* tab lists the seven constants with their uncertainty, the newest configuration and the change, the rotation, tilt and scales, and per file the lines used and the rms; the *Data* tab shows every line with the fitted edges (orange).

Accept the fit when:

check	expected
edge rms	a few µm (≤ ~10 µm)
lines left out	few, and only lines that do not cross the frame or do not see it
tilt of the gold face	≤ 2°
ST / SZ scales	within 0.5 % of 1
fitted edges (Data tab)	on the gold edges of every line
change from the newest configuration	plausible: same order as the known drifts; large changes explained (re-homing, optics)
verification (step 7)	the known sample found within about 0.05 mm





If something goes wrong

symptom	likely cause	action
the overview does not show the whole frame	frame not where expected (re-homing, mounting)	re-centre on the camera, repeat step 1
lines of group C miss the diamond	the newest configuration is more than ~1 mm off	fit groups A and B (with the overview) first, save it, make the plan again, measure group C with the new plan
a line flagged out of range / collision in the plan	frame far from the centre, or limits	check the frame position entered; change the line only if needed, the fit uses whatever lines are measured
many lines left out in the fit	wrong SN gold or gold height, files from another position	check SN gold, *h*, the file list
edge rms ≫ 10 µm	noisy data, sample drift, wrong motor read-back	look at the Data tab; repeat the worst setting
tilt > 2°	frame glued tilted	acceptable if the fit is good; the tilt is fitted
verification off by the same shift for several samples	frame not glued at the assumed place	measure a plate feature (riser edge) or a sample of known position and correct the frame position, then fit again

Limits of the method. The frame is the plate reference: its mounting offset cannot be seen by the fit. SN_f also takes up a beam passing beside the analyzer focus (by √2 × the offset); the found sample is then on the beam, not exactly at the analyzer focus. The ST / SZ scales are reported, not used by QuickAlignment.

Logbook sheet

Date: _____ · Optic: capillary / zone plate · Plate: _____ · People: _____

quantity	value	file / note
frame position assumed (x, y mm)		
SN_gold (step 2)		
SN peak on the riser		
gold height h = difference		
SN peak width (analyzer acceptance)		
optic position focused on the gold		
overview map file		
configuration used for the plan		

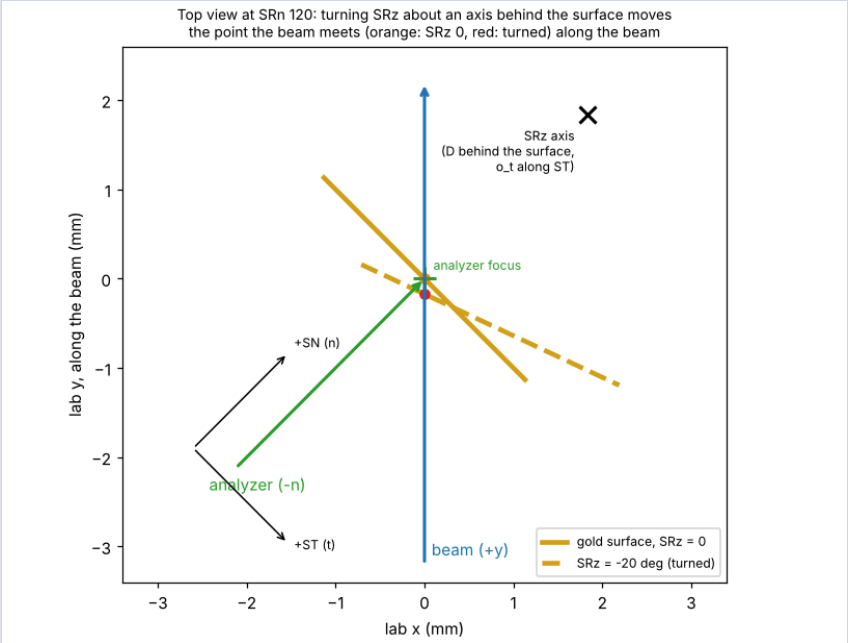
setting	SRn	SRz	SN set	file(s)	start time	remarks
S1	120	0				
S2	75	0				
S3	165	0				
S4	165	−25				
S5	165	−18				
S6	165	−11				
S7	165	−4				
S8	165	+3				
S9	165	+10				

result	value
edge rms / lines used	
ST0, SZ0, SN_f	
SRn axis (x, y)	
SRz offsets (ST, depth)	
tilt, ST scale, SZ scale	
saved configuration file	
verification sample, rotation, deviation	

Background: why SZ lines over a diamond

SZ is the only fast motor, so every measurement is an SZ line. At SRn 120 the frame's edges are horizontal and vertical, and an SZ line sees nothing of the horizontal position. Turned by 45° (SRn 165 or 75) the frame is a diamond: where an SZ line crosses its diagonal edges moves 1:1 with where the line

passes across it, so ST0, the SRn axis and the SRz axis are read on the fast SZ axis with full sensitivity. An SZ line lies in the surface and along the SRz axis, so the lit point does not slide along the beam and the analyzer acceptance does not move the edges.



Method, simulations of the perfect and imperfect cases and the accuracy: *Gold-frame calibration — Report* (`GoldFrameCalibration/docs/pdf/GoldFrame_calibration_report_en.pdf`).